Question Paper Code: 25901236

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECECT201 - PCB DESIGN AND 3DPRINTING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

ANSWER KEY

Part A

MCQ

Answer all questions

Each question carries **1** marks

1.	In Through-Hole Technology (THT), component leads are: b). Inserted into drilled holes	[U] / [CO1]
2.	Which of the following is an electrical design rule? b). Trace width and spacing	[U] / [CO1]
3.	Layers in multilayer PCB can go up to: d). 40	[U] / [CO1]
4.	The unit of inductance is: b). Henry	[U] / [CO1]
5.	Altium supports 3D view of PCB: a). Yes	[U] / [CO1]
6.	Proteus is mostly used before: b). Manufacturing	[U] / [CO1]
7.	In a schematic, wires are also called: c). Nets	[U] / [CO1]
8.	EAGLE stands for: b). Electronic Automation Graphic Layout Editor	[U] / [CO1]
9.	Which component is commonly used for manual ON/OFF control in a circuit? c). Switch	[U] / [CO1]
10.	Who is credited with developing the first Printed Circuit Board? c). Paul Eisler	[U] / [CO1]
11.	Which part of the PCB provides mechanical support? c). Substrate	[U] / [CO1]
12.	Compared to single sided PCB, double sided PCB is: b). More complex	[U] / [CO1]

13.	Current sense resistors usually have resistance values: b). Less than 1 ohm	[U] / [CO1]
14.	Vias are mainly used in: d). Both B and C (Double-sided PCB and Multilayer PCB)	[U] / [CO1]
15.	In a single sided PCB, components are placed on: a). One side only	[U] / [CO1]
16.	EDA software helps in drawing: a). Circuit diagrams	[U] / [CO1]
17.	After schematic capture, the next main step is: d). Board layout	[U] / [CO1]
18.	PCB prototypes are usually made to: d). Test and verify the designbefore mass production	[U] / [CO1]
19.	Gerber files are used for: c). PCB fabrication	[U] / [CO1]
20.	Electronic components are mainly classified into: b). Active and passive	[U] / [CO1]
21.	One disadvantage of FDM printing is: c). Visible layer lines	[U] / [CO2]
22.	Infill density mainly affects the: d). Strength and weight of theobject	[U] / [CO2]
23.	TinkerCad is mainly accessed through: c). Web browser	[U] / [CO2]
24.	Which CAD software is beginner-friendly and web-based? d). TinkerCAD	[U] / [CO2]
25.	The purpose of a gear mechanism project is to test: a). Motion and alignment	[U] / [CO2]
26.	Which device is required to use CAD software? c). Computer	[U] / [CO2]
27.	Selective Laser Sintering (SLS) works by: c). Fusing powder with laser	[U] / [CO2]
28.	For SLA printing, the minimum wall thickness is typically: a). 0.5–1 mm	[U] / [CO2]
29.	Which printing method is the most expensive among FDM, SLA, and SLS? c). SLS	[U] / [CO2]
30.	Designing for functionality includes considering: a). Tolerances and fit	[U] / [CO2]
31.	Fusion 360 can be used for:	[U] / [CO2]

c). CAD, CAM, and CAE

32. Which software is web-based and easy to learn? [U] / [CO2]

a). TinkerCad

33. Overhang angles greater than ____ usually require supports. [U] / [CO2]

c). 45°

34. In FDM printing, material is deposited: [U] / [CO2]

d). Layer by layer

35. What is 3D printing also known as? [U] / [CO2]

a). Additive manufacturing

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. Describe the key technical challenges involved in system integration projects. [An] / [CO3]

- Compatibility Issues: Different systems use different platforms, protocols, and data formats.
- Data Integration Problems: Inconsistent, duplicate, or missing data across systems.
- Legacy System Integration: Older systems lack APIs or modern interfaces.
- Scalability & Performance: Integrated system may slow down under high load.
- Security Risks: Increased vulnerabilities due to multiple interconnected systems.

Technical challenges mainly arise due to heterogeneity and require careful architecture design and testing.

37. Identify operational challenges during integration implementation and provide solutions. [An] / [CO3]

Challenges:

- Downtime during deployment
- User resistance to new system
- Training requirements
- Process disruption
- Lack of coordination between teams

Solutions:

- Use phased or parallel implementation to reduce downtime
- Provide user training and support programs
- Establish clear communication plans
- Use standard operating procedures (SOPs)
- Assign integration managers for coordination

38. Discuss the major stakeholder management challenges in integration initiativesand suggest suitable solutions. [An] / [CO3]

Challenges:

- Conflicting stakeholder requirements
- Lack of communication
- Resistance to change
- Unclear roles and responsibilities
- Expectation mismatch

Solutions:

- Conduct regular stakeholder meetings
- Define clear roles (RACI matrix)
- Use requirement documentation and validation
- Implement change management strategies
- Maintain transparent communication

39. Describe in detail a Smart City integration case study and explain its major components and outcomes. [An] / [CO3]

Example: Smart Traffic & Infrastructure System

Components:

- IoT Sensors: Traffic, pollution, and energy monitoring
- Communication Network: 4G/5G, fiber optics
- Data Integration Platform: Central control system
- Analytics & AI: Traffic prediction, energy optimization
- User Interfaces: Mobile apps, dashboards

Outcomes:

- Reduced traffic congestion
- Improved public safety
- Efficient energy usage
- Better urban planning
- Enhanced citizen services

Smart city integration combines IoT, data analytics, and communication systems to improve urban efficiency and quality of life.

This is only one example. Students may write different examples, so marks may provided accordingly.

40. Analyze a real-world example of integration in the education sector and itsimpact. [An] / [CO3]

Google Classroom Integration with Learning Tools

Many schools and colleges integrate Google Classroom with tools like Google Drive, Zoom, and Google Meet to create a unified digital learning environment.

Key Integration Features:

- Centralized Learning Platform: Assignments, notes, and announcements in one place
- Cloud Storage Integration: Easy sharing via Google Drive
- Live Classes: Conducted using Google Meet or Zoom
- Automated Assessment: Quizzes and grading tools
- Real-time Communication: Teacher-student interaction

Impact:

- Improved Accessibility: Students can learn anytime, anywhere
- Enhanced Collaboration: Easy sharing and group work
- Time Efficiency: Automated grading and submissions
- Continuity of Education: Especially during disruptions like COVID-19 pandemic
- Better Monitoring: Teachers can track student progress

Integration of digital platforms in education improves efficiency, accessibility, and quality of learning, making education more flexible and student-centered.

This is only one example. Students may write different examples, so marks may provided accordingly.

[5x3 = 15]